

The light propagates directly forward:

- ❑ Undiffracted beam; In phase; The central bright spot.

The path length difference between the rays from top and bottom is one wavelength:

- ❑ The irradiance drops to zero;

$$\sin \theta_1 = \pm \lambda / b$$

The path difference equals to two wavelength:

- ❑ The irradiance drops to zero $I(\theta_2) = 0$;

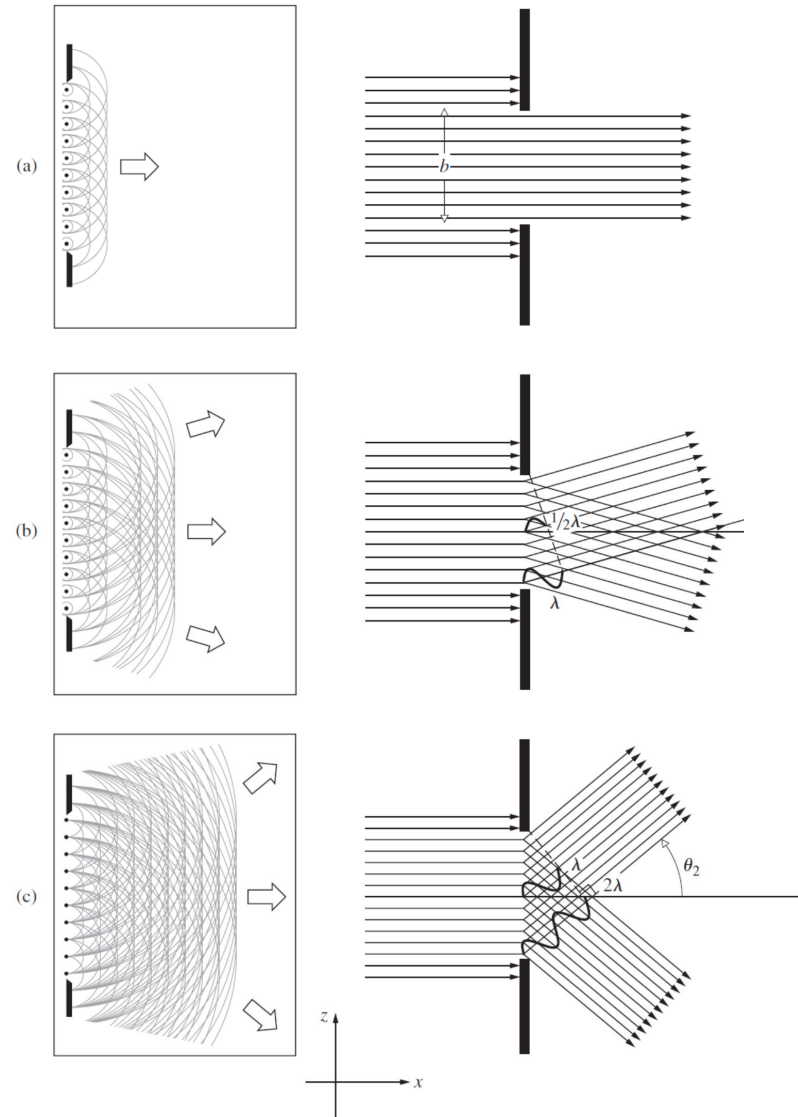
$$\sin \theta_2 = \pm 2 \lambda / b$$

In general:

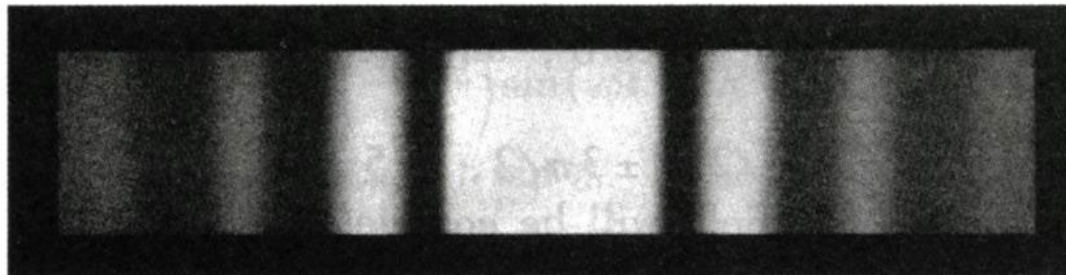
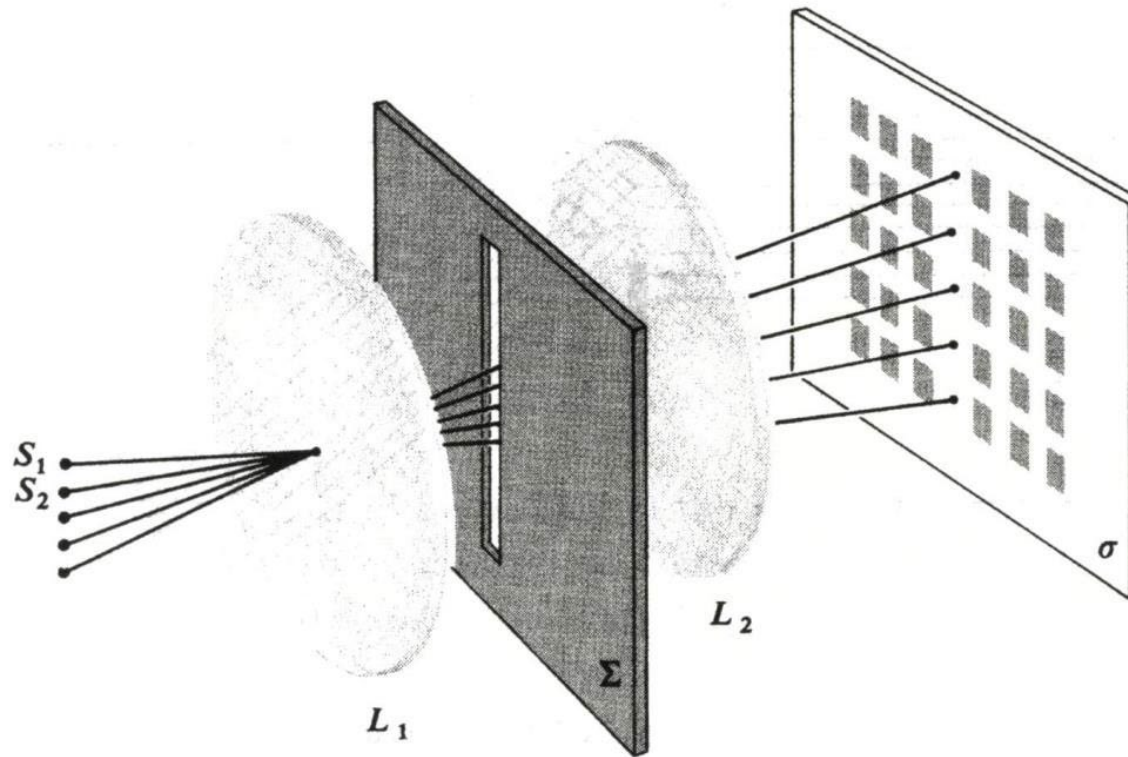
$$b \sin \theta_m = m \lambda$$

$$m = \pm 1, \pm 2, \pm 3, \dots$$

which is equal to minimum discussed above.

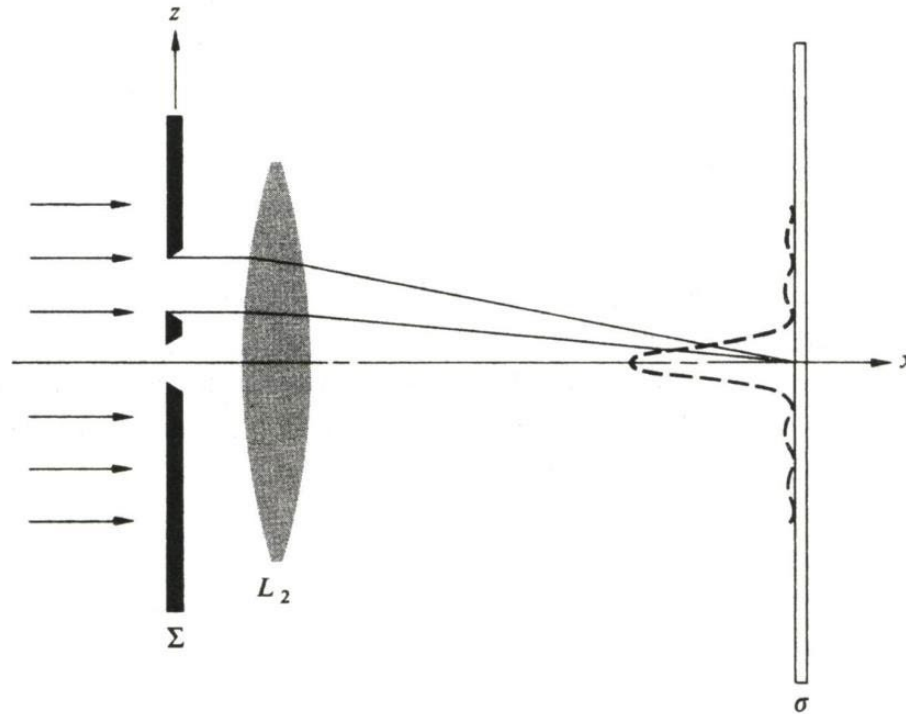


An incoherent line source positioned parallel to the slit.



The Double Slit

The diffraction pattern is centered about the axis of the lens, regardless of the slit's position



The double slit diffraction:

- ☐ Each aperture generate single-slit diffraction pattern.
- ☐ Two slits will interfere, which is determined by the differences of optical path lengths traversed by the overlapping wavelets from the two slits.

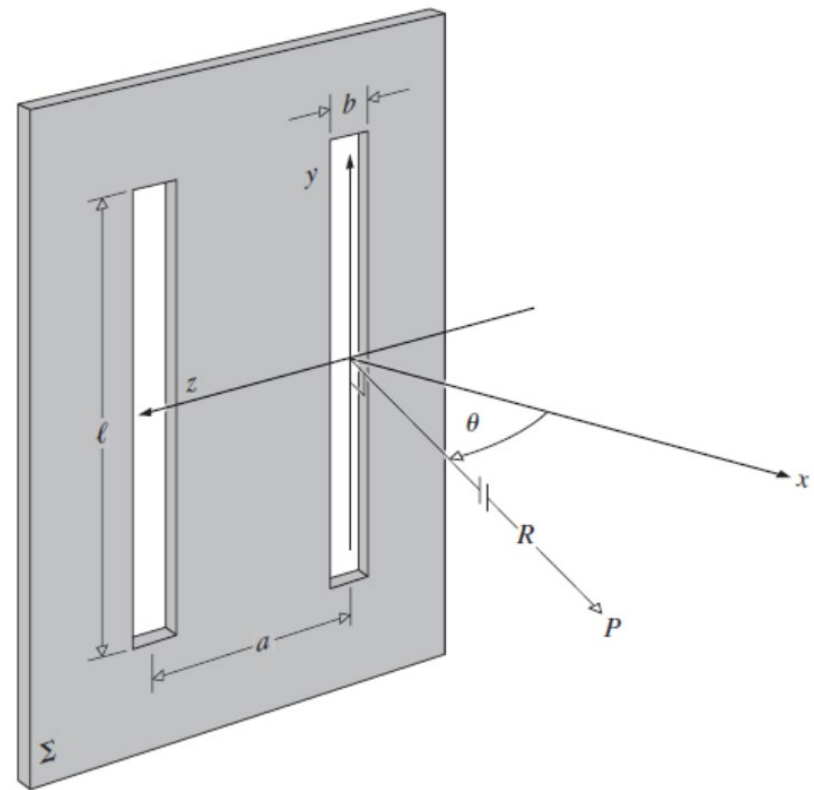
The expression of optical disturbance at a point on screen for double slit diffraction:

$$E = C \int_{-\frac{b}{2}}^{\frac{b}{2}} F(z) dz + C \int_{a-\frac{b}{2}}^{a+\frac{b}{2}} F(z) dz$$

$$F(z) = \sin[\omega t - k(R - z \sin \theta)]$$

$$E = bC \left(\frac{\sin \beta}{\beta} \right) [\sin(\omega t - kR) + \sin(\omega t - kR + 2\alpha)]$$

$$\beta \equiv \left(\frac{kb}{2} \right) \sin \theta \quad \alpha \equiv \left(\frac{ka}{2} \right) \sin \theta$$



This is just the sum of the two fields at P, one from each slit.

$$E = 2bC \left(\frac{\sin \beta}{\beta} \right) \cos \alpha \sin(\omega t - kR + \alpha)$$

$$I(\theta) = 2b^2 C^2 \left(\frac{\sin^2 \beta}{\beta^2} \right) \cos^2 \alpha = 4I_0 \left(\frac{\sin^2 \beta}{\beta^2} \right) \cos^2 \alpha$$

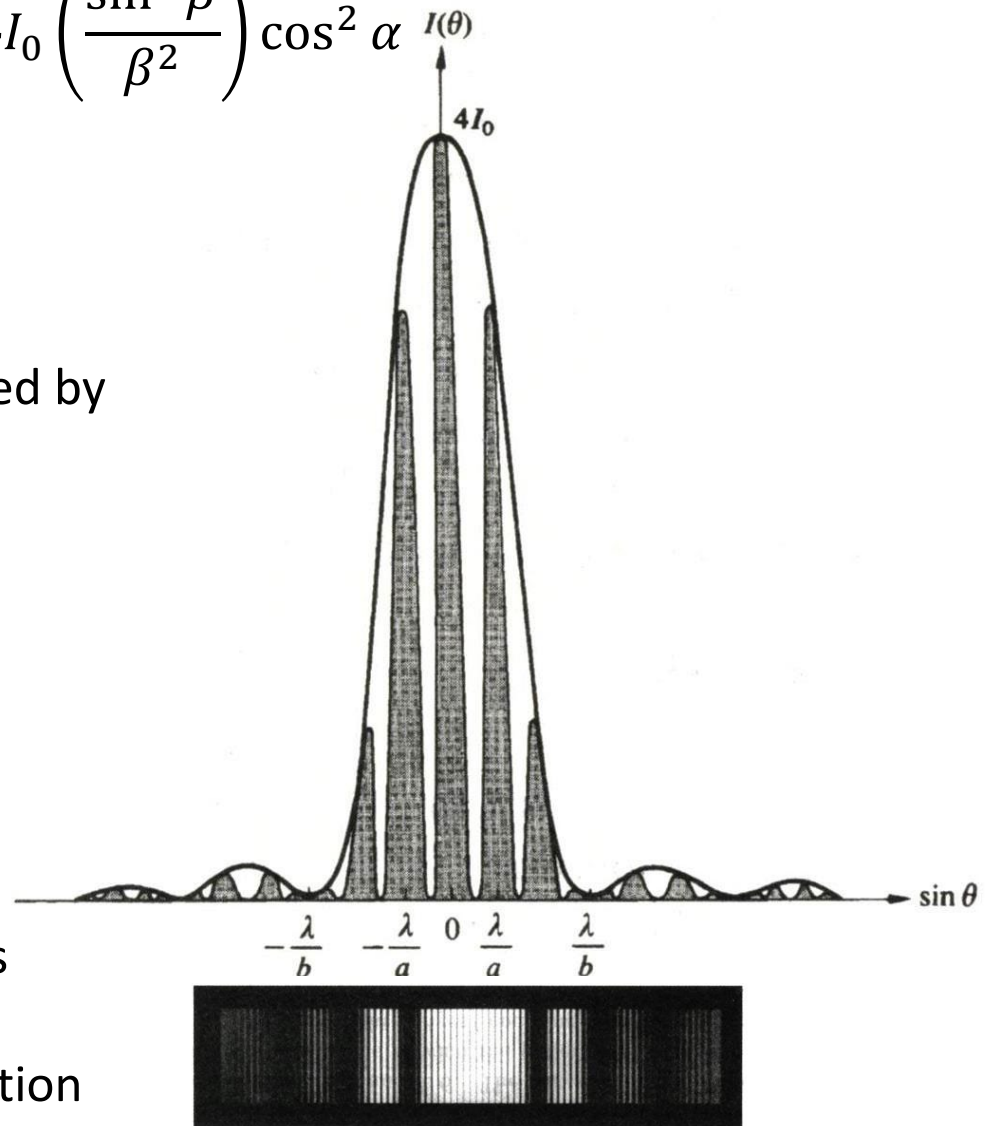
$$\beta \equiv \left(\frac{kb}{2} \right) \sin \theta \quad \alpha \equiv \left(\frac{ka}{2} \right) \sin \theta$$

A $\cos^2 \alpha$ interference term modulated by a $\sin^2 \beta / \beta^2$ diffraction term.

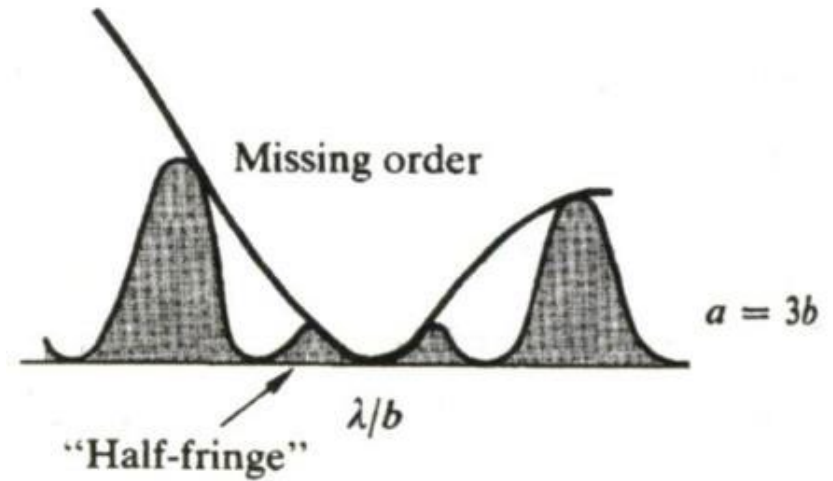
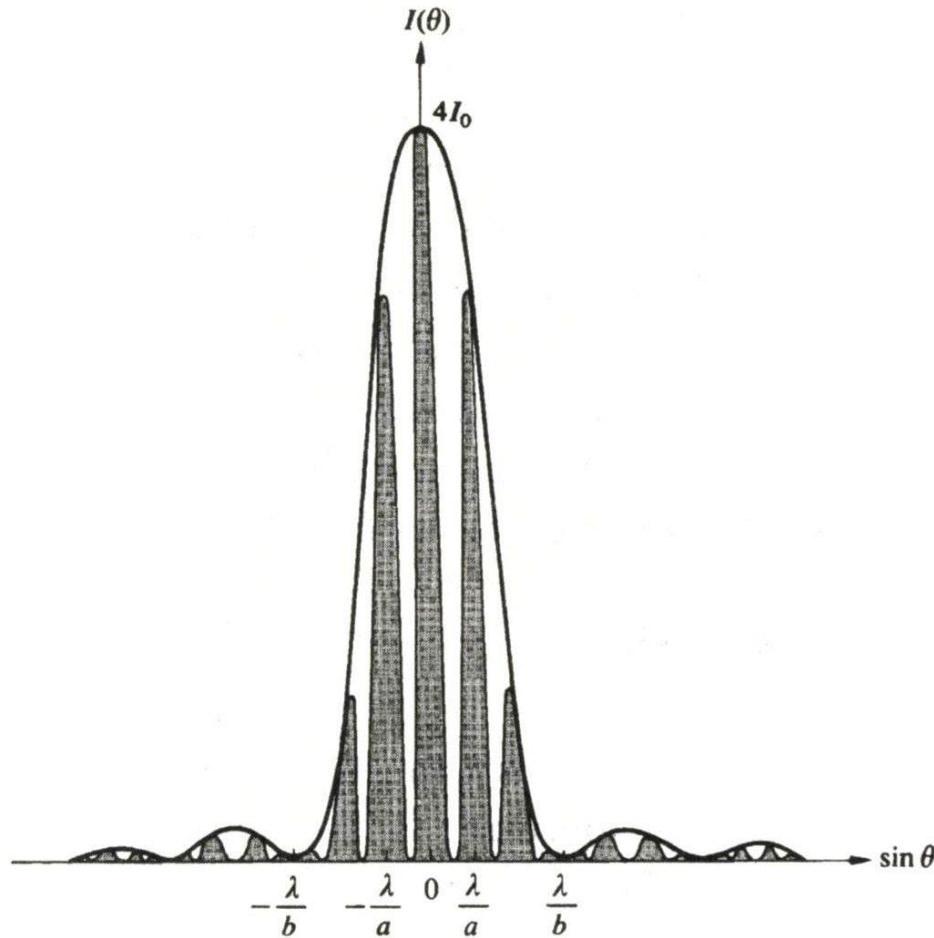
$I(\theta) = 0$, when

$$\beta = \pm\pi, \pm2\pi, \pm3\pi, \dots$$

$$\alpha = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \dots$$



- (I) $kb \ll 1$, then $(\sin \beta)/\beta \approx 1$: Young's experiment
- (II) $a=0$, then $\alpha=0$: single-slit diffraction



Missing order: an interference maximum and a diffraction minimum occur at the same θ -value.

Example: when $a=mb$

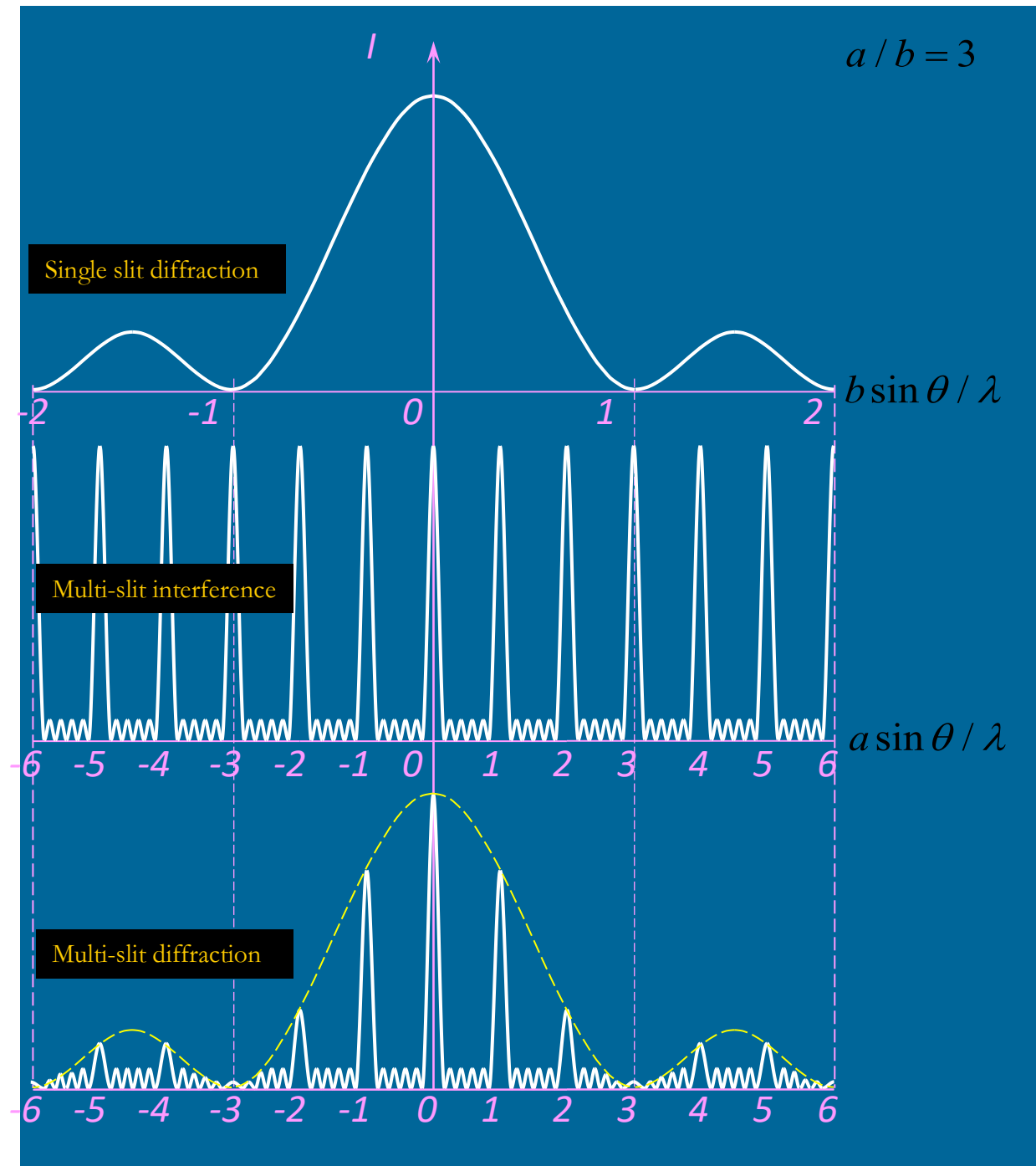
$$\text{Number of fringes: } N = \frac{\frac{2\lambda}{b}}{\frac{\lambda}{a}} = \frac{2a}{b} = 2m$$

There is $2m$ bright fringes within the central diffraction peak.

$$I(\theta) = 4I_0 \left(\frac{\sin^2 \beta}{\beta^2} \right) \cos^2 \alpha$$

$$\beta \equiv \left(\frac{kb}{2} \right) \sin \theta$$

$$\alpha \equiv \left(\frac{ka}{2} \right) \sin \theta$$



Diffraction by Many Slits

Diffraction by N slits with the width b and center-to-center separation a .

$$\begin{aligned}
 E = & C \int_{-\frac{b}{2}}^{\frac{b}{2}} F(z) dz + C \int_{a-\frac{b}{2}}^{a+\frac{b}{2}} F(z) dz \\
 & + C \int_{2a-\frac{b}{2}}^{2a+\frac{b}{2}} F(z) dz + \dots \\
 & + C \int_{(N-1)a-\frac{b}{2}}^{(N-1)a+\frac{b}{2}} F(z) dz
 \end{aligned}$$

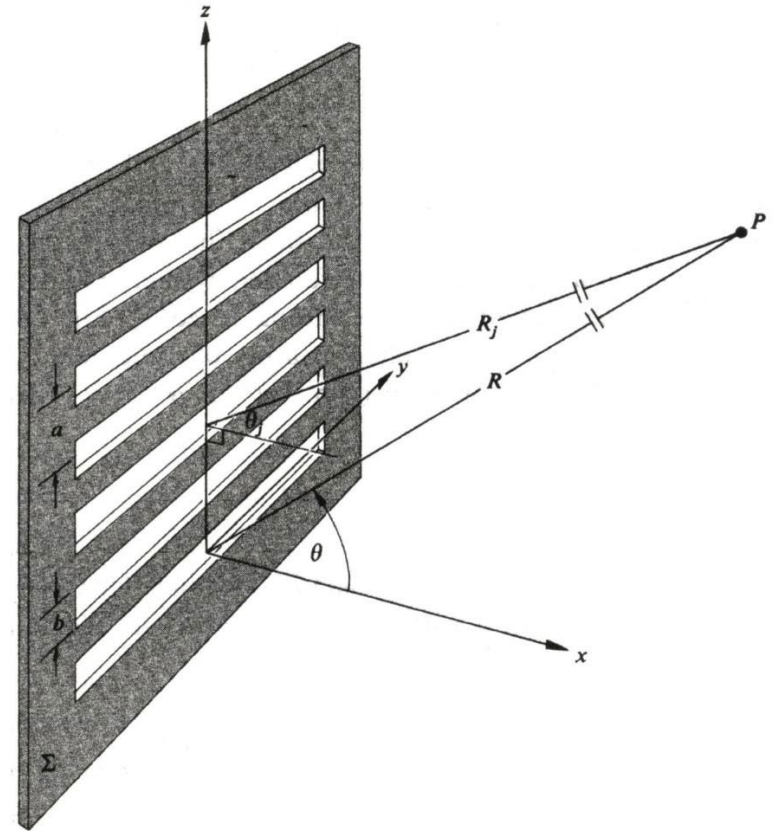
$$F(z) = \sin[\omega t - k(R - z \sin \theta)]$$

Fraunhofer condition: $r = R - z \sin \theta$

Require $\theta_j \approx \theta$

$$E = bC \left(\frac{\sin \beta}{\beta} \right) \left(\frac{\sin N\alpha}{\sin \alpha} \right) \sin[\omega t - kR + (N-1)\alpha]$$

$R - (N-1) \left(\frac{a}{2} \right) \sin \theta$: the distance from the center of the array to the point P.



The flux-density distribution function:

$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$$

I_0 is the flux density on the $\theta = 0$ direction emitted by any one of the slits; $I(0) = N^2 I_0$.

A multiple-wave interference

$\left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$ system modulated by the single-slit diffraction envelop

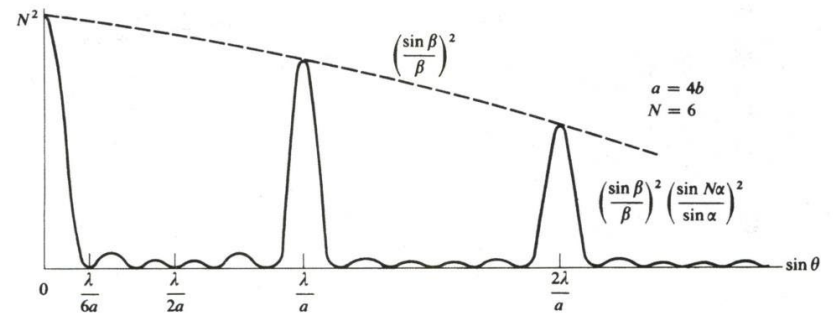
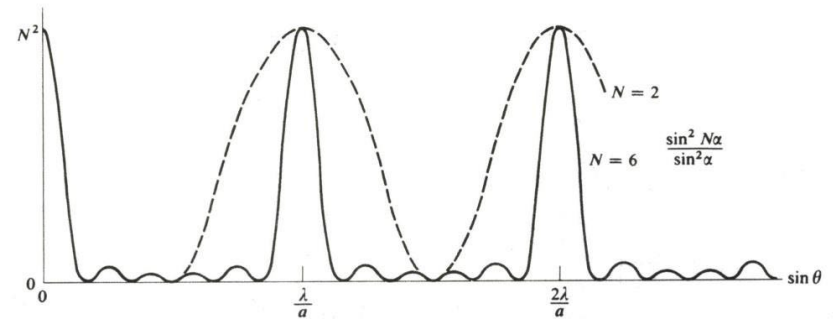
$$\left(\frac{\sin \beta}{\beta} \right)^2$$

$$N = 1, \quad I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin \alpha}{\sin \alpha} \right)^2 = I_0 \left(\frac{\sin \beta}{\beta} \right)^2$$

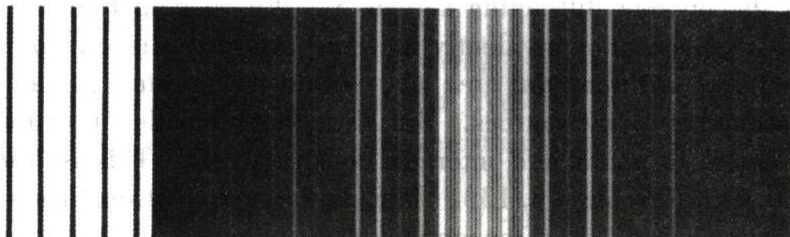
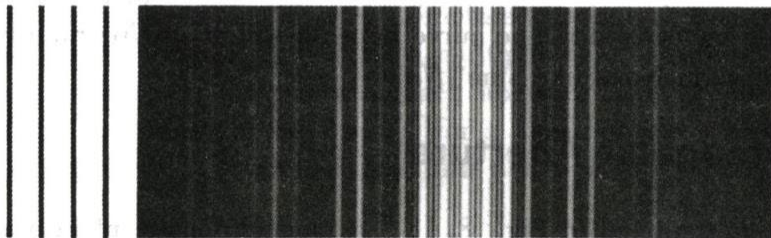
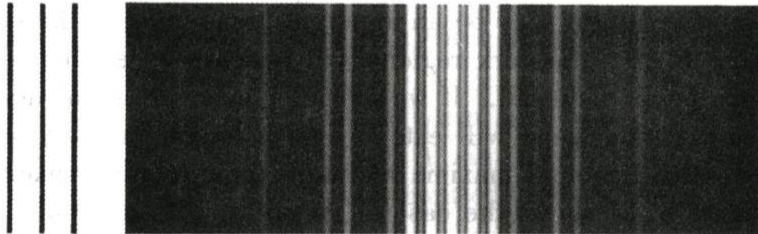
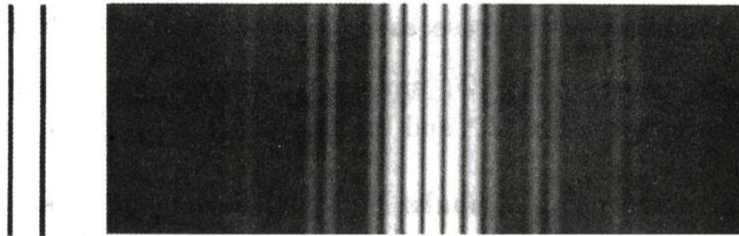
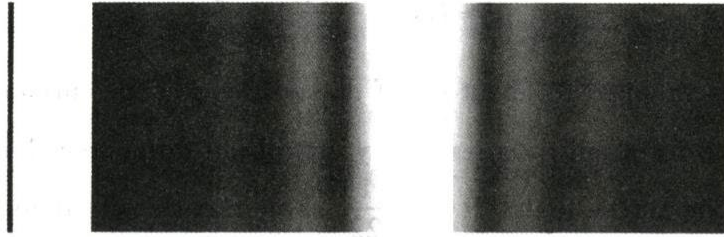
single slit diffraction formula

$$N = 2, \quad I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin 2\alpha}{\sin \alpha} \right)^2 = 4I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \cos^2 \alpha$$

double slit diffraction formula



Between two adjacent principle irradiance maxima, there are $(N-1)$ minima and $(N-2)$ subsidiary maxima.



The flux-density distribution function:

$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$$

Between two adjacent principal irradiance maxima, there are $(N-1)$ minima and $(N-2)$ subsidiary maxima.

Homework

□ 10.2, 10.10, 10.12

The Diffraction Grating

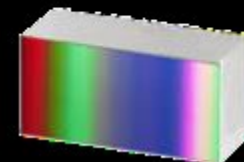
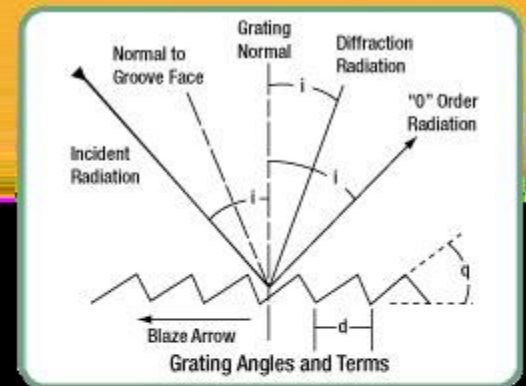
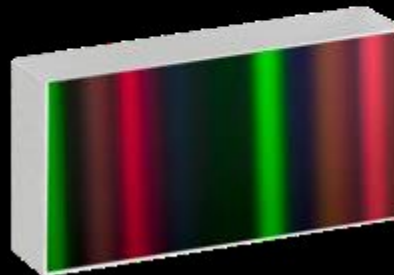
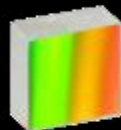
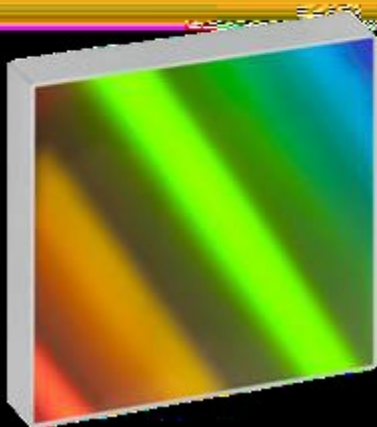
A repetitive array of diffracting elements, either apertures or obstacles, that has the effect of producing periodic alterations in the phase, amplitude, or both of an emergent wave.

Reflection phase grating

300 nm to 10.6 μm Blaze Wavelengths

Higher Efficiencies than Holographic Gratings

5 Sizes Available with Sides Measuring from 1/2" to 2"



Recall: **diffraction by many slits**

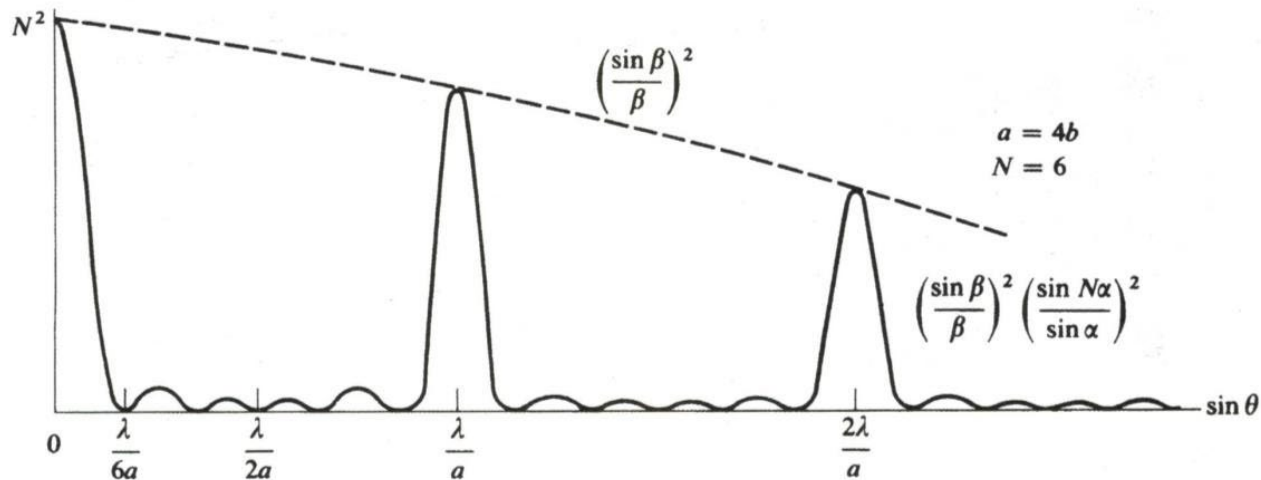
Irradiance at P:
$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$$

$$\beta \equiv \left(\frac{kb}{2} \right) \sin \theta \quad \alpha \equiv \left(\frac{ka}{2} \right) \sin \theta$$

Principal maxima: $\sin N\alpha = 0$ and $\sin \alpha = 0$

$$\alpha = 0, \pm\pi, \pm2\pi, \dots \pm m\pi$$

$$a \sin \theta_m = m\lambda$$

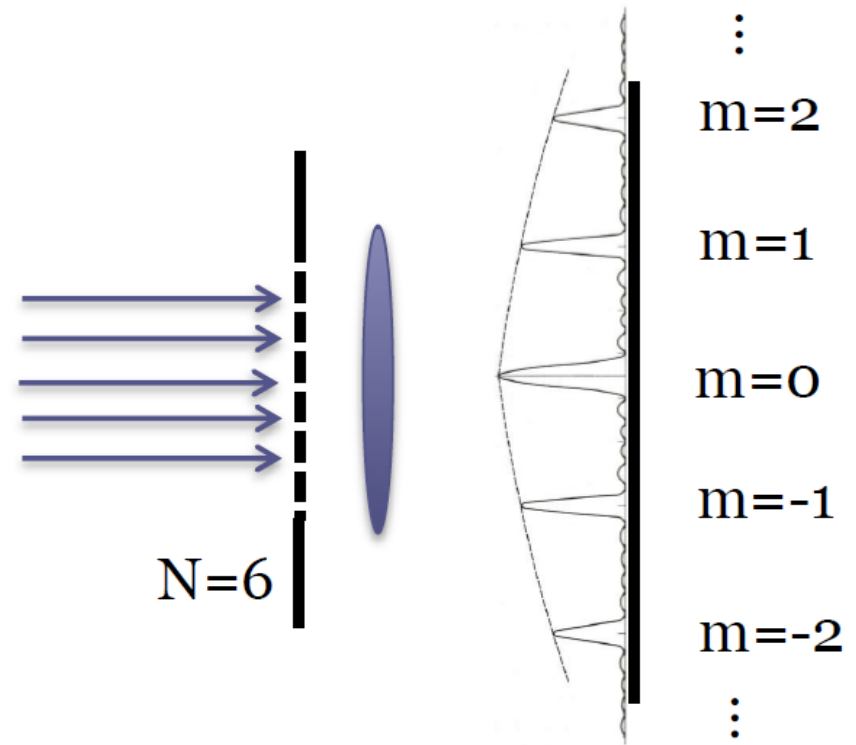
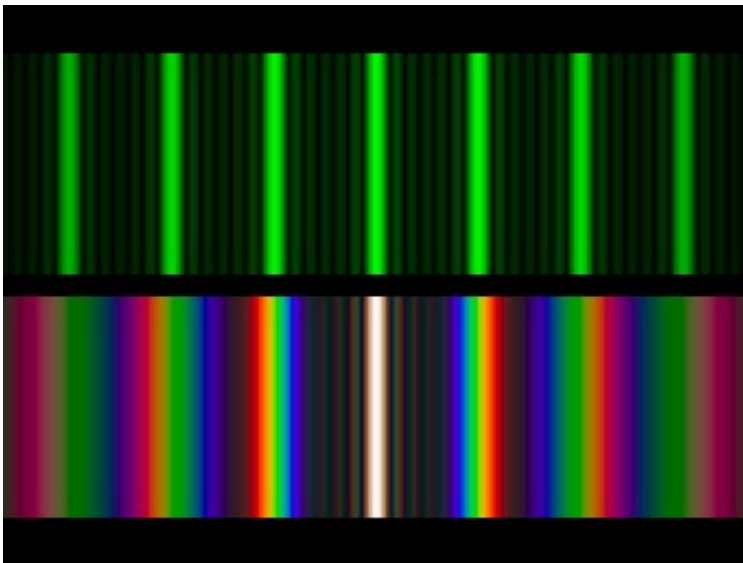


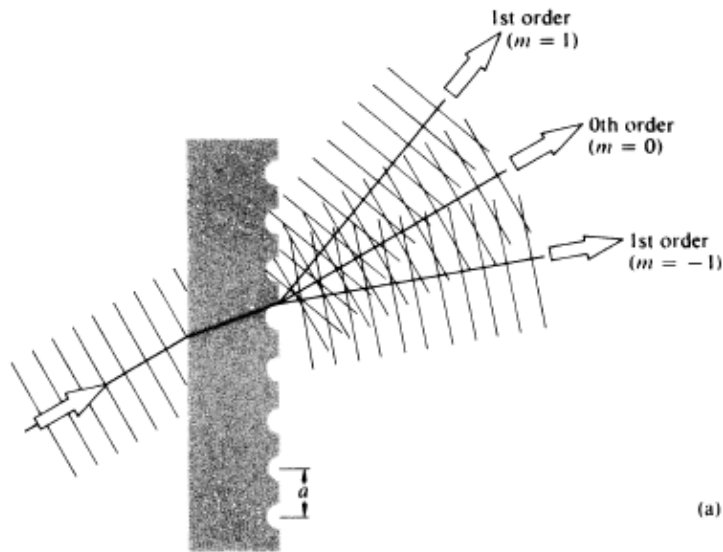
Grating equation for normal incidence:

$$a \sin \theta_m = m\lambda$$

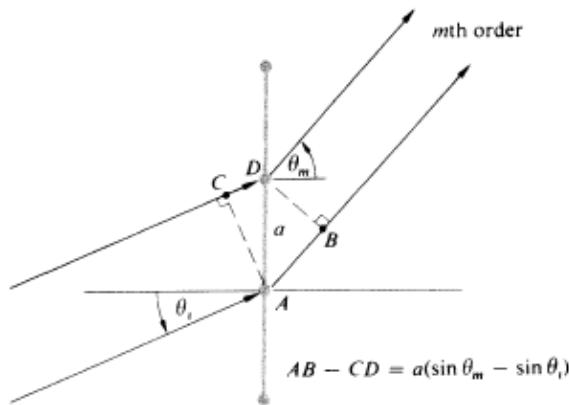
For $m=0$, $\theta_0=0$: image corresponds to the undeflected, white-light view of the source.

For $m \neq 0$, the grating equation is dependent on λ : the various colored images of the source corresponding to slightly different angles.

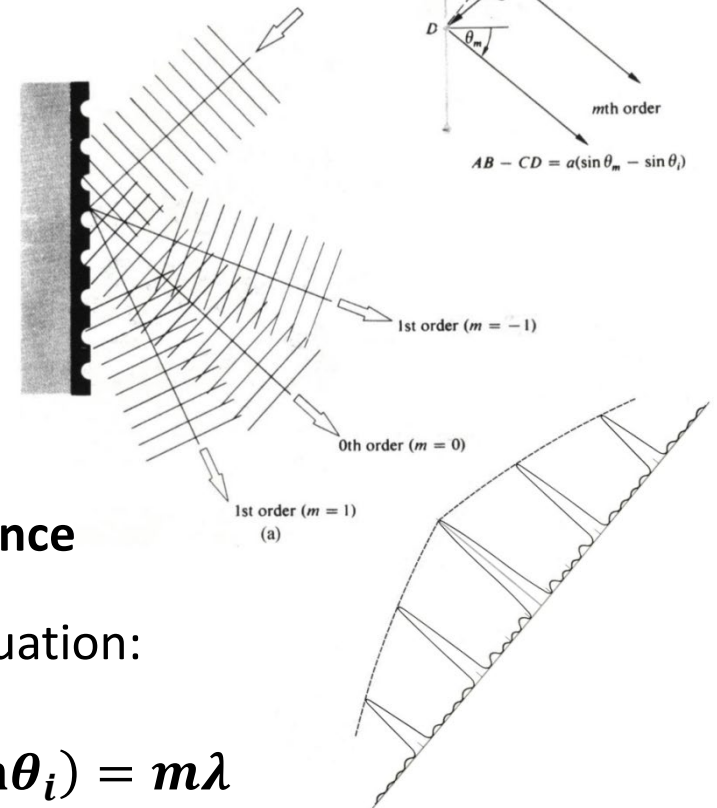




(a)



(b)



Oblique incidence

The grating equation:

$$a(\sin\theta_m - \sin\theta_i) = m\lambda$$

The zeroth order: $\theta_m = \theta_i$.

For spectroscopic purposes, all of the light is essentially wasted.

Idea: shift energy out of the useless zeroth order into one of the higher-order spectra.

Blazed reflection phase grating:

Recall

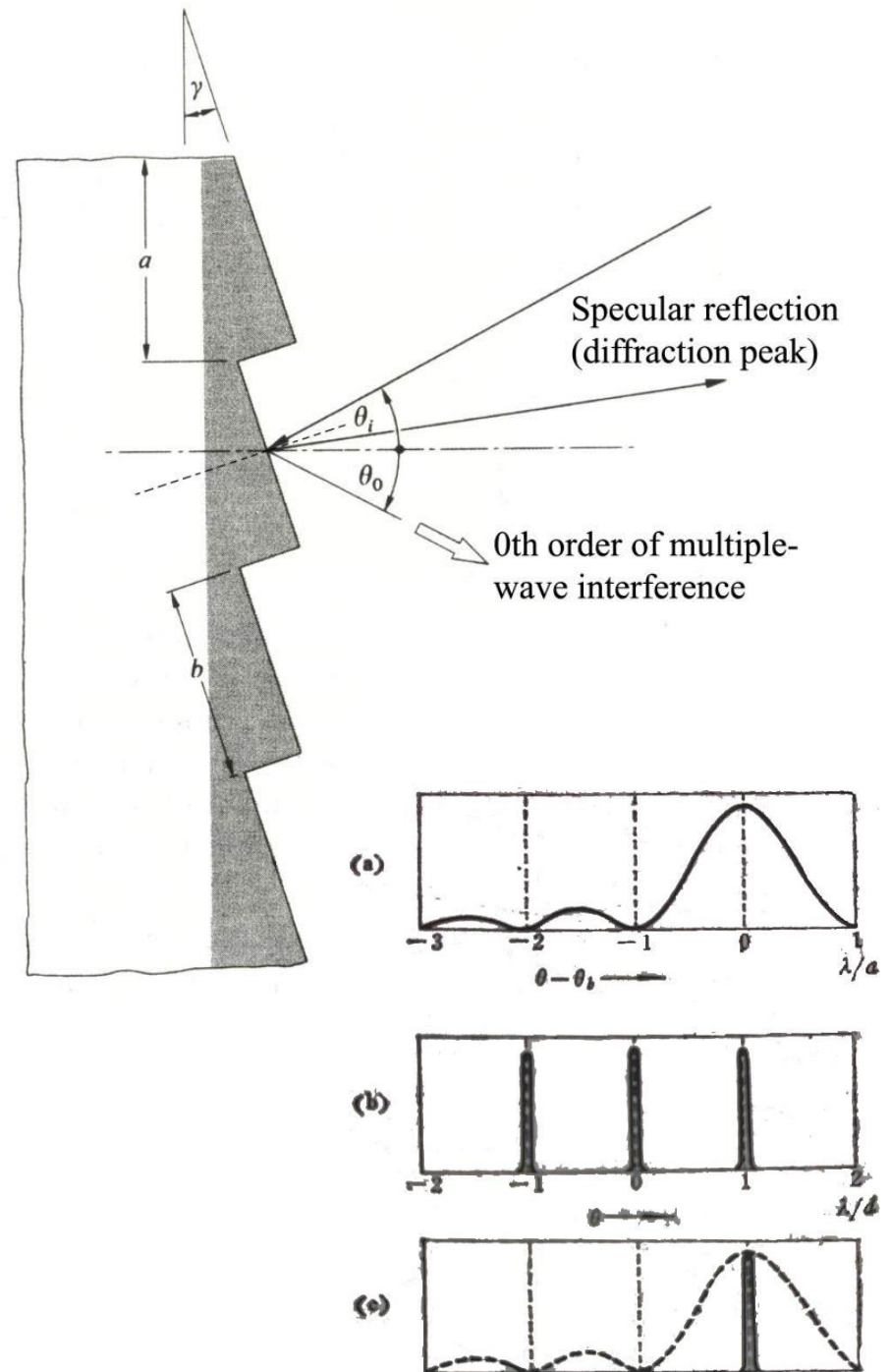
$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$$

Single-facet diffraction

$\left(\frac{\sin \beta}{\beta} \right)^2$ corresponds to specular reflection off that face, for each groove, which is governed by the blaze angle γ .

A multiple-wave interference

$\left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$ or θ_m value are determined by a , λ and θ_i . But θ_m and θ_i are measured from the normal to the grating plane and not with respect to the individual groove surfaces.



Example: the normal incidence

Single-facet diffraction

$$\left(\frac{\sin \beta}{\beta}\right)^2 :$$

$$\theta_i - \theta_r = 2\gamma \rightarrow \theta_r = -2\gamma$$

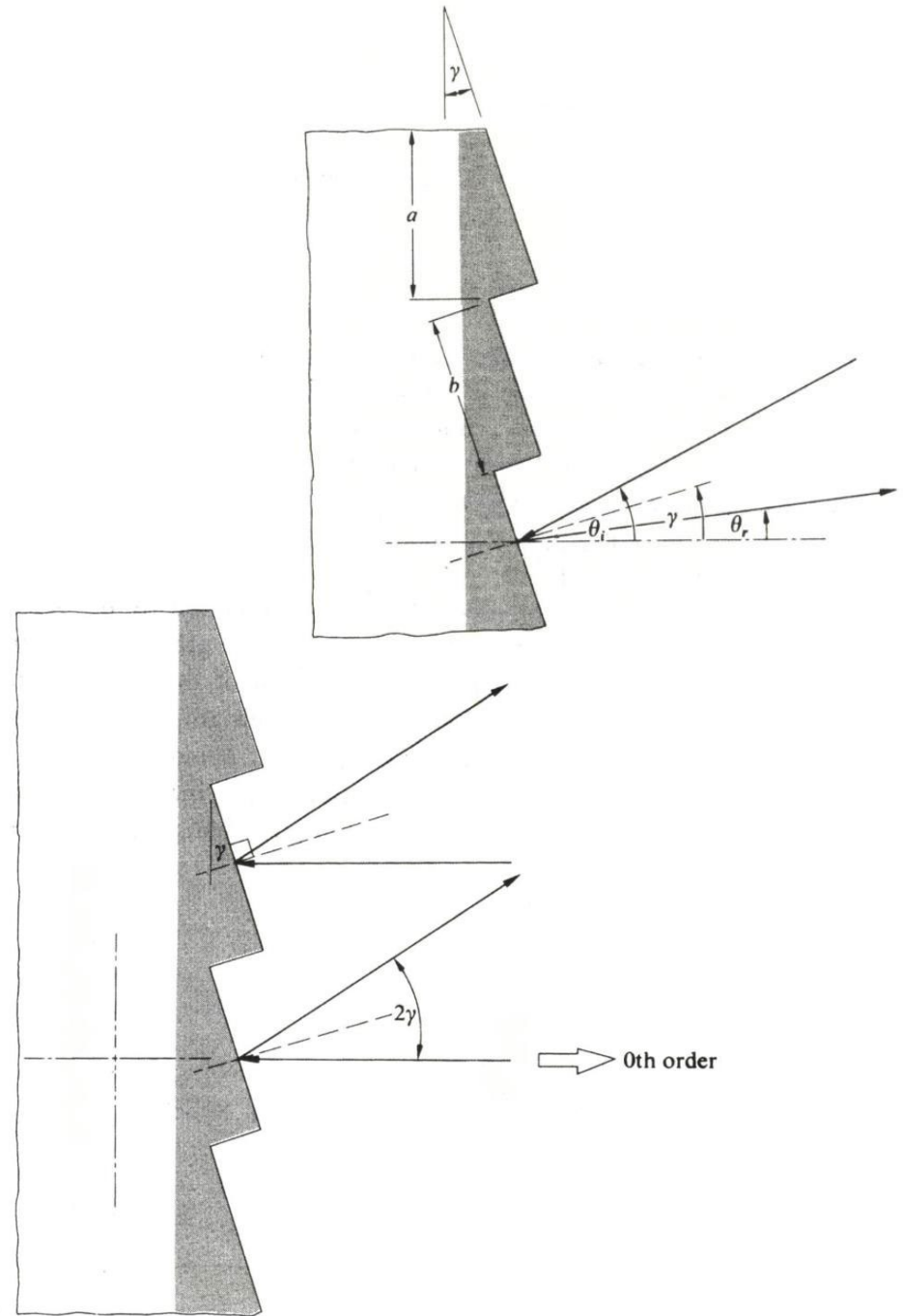
A multiple-wave interference term

$$\left(\frac{\sin N\alpha}{\sin \alpha}\right)^2 : \theta_i = 0$$

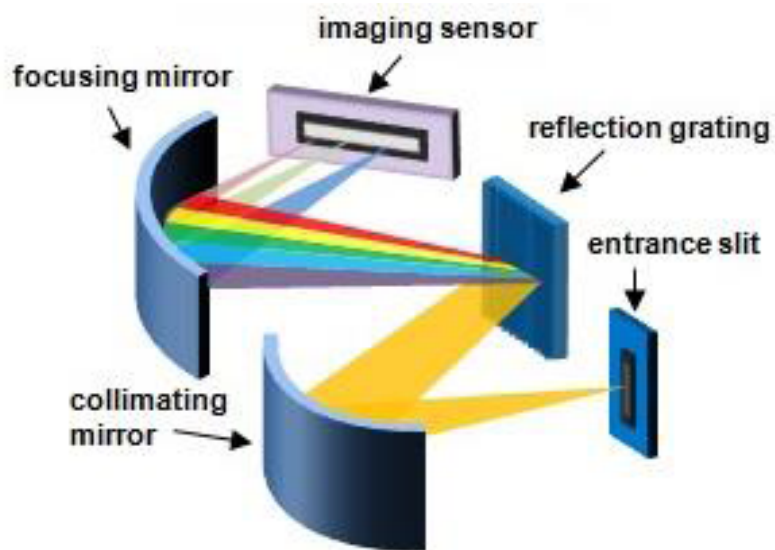
$$\text{For } m = 0 ; \theta_0 = 0$$

Principle maxima:

$$a \sin(-2\gamma) = m\lambda$$



Spectrometer



The Arbitrary Aperture

Huygens-Fresnel Principle:

A differential area dS : it is very small, so that *all the contributions at P* are in phase; it emits a spherical wave.

The optical disturbance at P:

$$dE = \left(\frac{\epsilon_A}{R} \right) e^{i(\omega t - kr)} dS$$

Fraunhofer condition, the total disturbance:

$$\tilde{E} = \frac{\epsilon_A e^{i(\omega t - kR)}}{R} \int_{\text{Aperture}} e^{ik(Yy + Zz)} dS$$

